

*Proof.* 1. **Degenerate cases.** If  $n = 1$  the statement reads  $a_1 b_1 \geq a_1 b_1$ , which holds with equality. If the  $a$ -sequence is constant, say  $a_i = c$  for all  $i$ , then

$$n \sum_{i=1}^n a_i b_i = nc \sum_{i=1}^n b_i = \left( \sum_{i=1}^n a_i \right) \left( \sum_{i=1}^n b_i \right),$$

so the inequality is an equality. The same argument applies when the  $b$ -sequence is constant. Hence we may assume  $n \geq 2$  and that neither sequence is identically constant.

2. **Pairwise non-negativity.** For any indices  $i < j$  we have  $a_i \leq a_j$  and  $b_i \leq b_j$ ; therefore

$$(a_i - a_j)(b_i - b_j) \geq 0. \quad (1)$$

3. **Summing over all pairs.** Summing (1) over all unordered pairs  $(i, j)$  with  $1 \leq i < j \leq n$  gives

$$\sum_{1 \leq i < j \leq n} (a_i - a_j)(b_i - b_j) \geq 0. \quad (2)$$

4. **Algebraic identity.** For a fixed pair  $(i, j)$ ,

$$(a_i - a_j)(b_i - b_j) = a_i b_i + a_j b_j - a_i b_j - a_j b_i.$$

Summing this expression over all  $i < j$  yields

$$\begin{aligned} \sum_{i < j} (a_i - a_j)(b_i - b_j) &= \underbrace{\sum_{i < j} (a_i b_i + a_j b_j)}_{\text{each } a_k b_k \text{ appears } n-1 \text{ times}} - \underbrace{\sum_{i < j} (a_i b_j + a_j b_i)}_{\text{each } a_k b_\ell \text{ (} k \neq \ell \text{) appears once}} \\ &= (n-1) \sum_{k=1}^n a_k b_k - \left( \sum_{k=1}^n a_k \right) \left( \sum_{\ell=1}^n b_\ell \right) + \sum_{k=1}^n a_k b_k \\ &= n \sum_{k=1}^n a_k b_k - \left( \sum_{k=1}^n a_k \right) \left( \sum_{\ell=1}^n b_\ell \right). \end{aligned} \quad (3)$$

The counting argument above is valid for every integer  $n \geq 2$ .

5. **Chebyshev's inequality.** Insert (3) into (2):

$$0 \leq n \sum_{i=1}^n a_i b_i - \left( \sum_{i=1}^n a_i \right) \left( \sum_{i=1}^n b_i \right).$$

Rearranging gives the desired inequality

$$n \sum_{i=1}^n a_i b_i \geq \left( \sum_{i=1}^n a_i \right) \left( \sum_{i=1}^n b_i \right). \quad (4)$$

6. **Equality condition.** Equality in (4) can occur only if equality holds in every instance of (1), because (2) is a sum of non-negative terms. For a particular pair  $(i, j)$  we have equality in (1) precisely when  $a_i = a_j$  or  $b_i = b_j$ . Thus equality in the whole sum (2) forces either

- $a_i = a_j$  for all  $i, j$  (the  $a$ -sequence is constant), or
- $b_i = b_j$  for all  $i, j$  (the  $b$ -sequence is constant).

The case  $n = 1$  is included in both possibilities. Hence equality holds *iff* one of the two sequences is constant.

7. **Conclusion.** For any integer  $n \geq 1$  and any real numbers

$$a_1 \leq a_2 \leq \cdots \leq a_n, \quad b_1 \leq b_2 \leq \cdots \leq b_n,$$

the inequality

$$n \sum_{i=1}^n a_i b_i \geq \left( \sum_{i=1}^n a_i \right) \left( \sum_{i=1}^n b_i \right)$$

holds, with equality exactly when the  $a$ -sequence or the  $b$ -sequence is constant (including the trivial case  $n = 1$ ).  $\square$

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